

heart and the roots of the major blood vessels emerging from it. The inner layer attaches to the heart wall.

Periosteum

The tough **tissue** that coats all bone surfaces except joints and from which new bone can be formed; contains blood and lymphatic vessels and **nerves**.

Peripheral nervous system

All the **nerves** that fan out from the brain and spinal cord, linking these parts with the rest of the body. The system consists of **cranial nerves** and **spinal nerves**.

Peristalsis

A coordinated succession of contractions and relaxations of the muscular wall of a tubular structure, such as the intestines, that moves the contents along.

Peritoneum

The double-layered membrane that lines the inner wall of the abdomen. The peritoneum covers and partly supports the abdominal **organs**. It also secretes a fluid that lubricates the movement of the intestines.

Phagocyte

A **white blood cell** or similar cell that surrounds and engulfs unwanted matter, such as invading microbes and cellular debris.

Pharynx

The passage leading down from the back of the nose and the mouth to the **oesophagus**; it consists of the nasopharynx, the oropharynx, and the laryngopharynx.

Pituitary gland

A pea-sized gland hanging from the underside of the brain. The pituitary secretes **hormones** that control many other glands in the body, and is regulated by the **hypothalamus**.

Placenta

The disc-shaped **organ** that forms in the **uterus** during pregnancy. It links the blood supplies of the mother and baby via the **umbilical cord** and nourishes the growing baby.

Plasma

The fluid part of the blood from which all cells have been removed; it is

mostly water, but contains some **proteins**, salts, and various nutrients, including glucose.

Platelet

Tiny fragment of a type of large cell manufactured in bone marrow and known as a megakaryocyte. Platelets are present in large numbers in the blood and are vital for blood clotting.

Pleura

A double-layered membrane, the inner layer of which covers the lung while the outer layer lines the chest cavity. A layer of fluid lubricates and enables movement between the two.

Progesterone

A female **sex hormone** secreted by the **ovaries** and **placenta** that allows the **uterus** to receive and retain a fertilized egg.

Prostaglandins

A group of fatty acids, made in the body, that have various functions and influence some hormones.

Prostate gland

A male accessory sex gland situated at the base of the bladder and opening into the **urethra**. It secretes some of the fluid in semen.

Protein

Huge molecule composed of chains of **amino acids**; the basis of many structural materials (keratin, **collagen**), **enzymes**, and **antibodies**.

Pulmonary artery

The large **artery** that conveys deoxygenated blood from the right **ventricle** of the heart to the lungs to be reoxygenated.

R

Red blood cells

Biconcave, disc-shaped cells, without **nuclei**, that contain **haemoglobin**. There are 4–5 million red cells in 1 millilitre ($\frac{1}{500}$ pints) of blood.

Renal

Relating to the **kidneys**.

Respiration

1. Bodily movements of breathing.
2. Gas exchange of oxygen for carbon dioxide in the lungs. 3. Similar

gas exchange in the **tissues** (cellular respiration). 4. Breakdown of molecules such as glucose to release their energy for cellular functions.

Retina

A light-sensitive layer lining the inside of the back of the eye; it converts optical images to **nerve** impulses, which travel to the brain via the **optic nerve**.

RNA

Ribonucleic acid, a substance present in cells; different forms carry out various functions, including the transfer of genetic information and manufacture of **proteins**.

S

Saliva

A watery fluid secreted into the mouth by the salivary glands to aid chewing, tasting, and digestion.

Sex hormones

Steroid substances, including **testosterone** in males and **oestrogen** and **progesterone** in females, that bring about the development of bodily sexual characteristics. Sex hormones also regulate sperm and egg cell production and the menstrual cycle.

Sphincter

A muscle ring, or local thickening of the muscle coat, surrounding an opening in the body, such as the anus or the **urethra**.

Spinal nerves

The 31 pairs of combined motor and sensory **nerves** that emerge from and enter the spinal cord.

Spine

The column of 33 ring-like bones, called **vertebrae**, that divides into seven cervical vertebrae, 12 thoracic vertebrae, five lumbar vertebrae, and the fused vertebrae of the sacrum and coccyx.

Spleen

A lymphatic **organ**, situated on the upper left of the abdomen, that destroys worn-out **red blood cells**, filters out impurities from the blood, and helps to fight infection.

Stem cell

Generalized type of cell, usually fast-dividing, with the potential to